



June 2, 2022

Samantha Murray, President
California Fish and Game Commission
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RE: Agenda Item 5: Petition to list the Western Joshua Tree as a Threatened Species in accordance with the California Endangered Species Act – Support “Threatened” Listing

Submitted via email to fgc@fgc.ca.gov

Dear President Murray and Commissioners:

Thank you for this opportunity to provide these comments regarding the petition to list the Western Joshua Tree as a Threatened species under the auspices of the California Endangered Species Act. I submit these comments in addition to my comments submitted to the California Department of Fish and Wildlife (CDFW) as one of their peer reviewers (Peer Review Comments attached).

The scientific data presented by the CDFW in their Status Review of Western Joshua Tree (*Yucca brevifolia*), dated March 2022, is very good and comprehensive. Yet, contrary to four out of the five peer reviewers, they come to the wrong conclusion when they summarily state that, “The Department recommends that the Commission find the petitioned action to list western Joshua tree as a threatened species to be not warranted.” This conclusion, limited to a very brief two paragraph statement, flies in the face of the facts the Status Review so painstakingly presents in the previous 125 pages.

My own peer review focused on the Department’s lack of consideration of the appropriateness of listing the WJT as Threatened in its Southern Population—the WJT distribution extending from Palmdale-Lancaster on the west to Joshua Tree National Park to the east (Fig. 4, pg. 17). The Southern Population of WJT should be considered an “Evolutionarily Significant Unit” unto itself, considered separately from the Northern Population extending from Lone Pine-Bishop, northwards along the Eastern Sierra range.

Within the Southern Population, a Threatened species designation, according to the definition under the CESA, is well warranted. The lower elevation portions of the WJT range in the south (northeast of Lancaster, extending to Barstow) already exhibit only 0-1% cover and very sparse distributions. The obligate pollinating moth, *Tegeticula synthetica*, is not present at these lower elevations (Webber 1953, Rowlands 1978, Harrower and Gilbert 2018), indicating that sexual reproduction (plants from seeds) is no longer occurring in this portion of the WJT range and the surviving plants are doomed within their generation lifetimes—within the

“foreseeable future” of the Year 2100. The intermediate elevations within the Southern Population have been highly fragmented and cleared within the cities of Palmdale-Lancaster, Hesperia-Adelanto-Victorville-Apple Valley, and Joshua Tree-Yucca Valley-Twenty-nine Palms. The remaining fragments of habitat across this broad sweep of WJT habitat have been so highly subdivided that the pollinating moths may no longer be able to sustain themselves in such small patches; and the prospects of seed dispersal (measured in meters) is nil. Finally, at the upper elevations of the Southern WJT range, the habitat has been decimated by increasingly large and frequent wildfires. Although the Department states that these wildfires remove just 2.5% of the overall WJT habitat per decade, if one considers these wildfires within just the higher elevation portions of the Southern WJT range, they easily represent 7.5-10% of remaining habitat per decade—such that more than half of this remaining habitat will have been burned by the year 2050. Recovery of these Joshua tree woodlands will require more than a century.

The Status Review itself summarizes the situation: “Considered collectively, the direct and indirect effects of climate change, the direct and indirect effects of [urban] development and other human activities, and the direct and indirect effects of wildfire are interconnected and will affect different portions [the Southern population] of western Joshua tree’s range in different ways [very severely], sometimes cumulatively.” [Insertions by me]

The Status Review continues with the summary that, “Climate change may reduce recruitment and abundance in southern and lower elevation portions of the [WJT] range, development and other human activities are expected to destroy and modify habitat [in the middle elevations of the Southern WJT range], and fire is expected to kill a [substantial] proportion of trees in [the higher elevation areas of the Southern WJT range].”

Together, climate change has already reduced or eliminated reproduction of WJTs at the lower elevations of their range. Urbanization has already developed and highly fragmented the middle elevations of the WJT range; and wildfires have decimated hundreds of thousands of acres of WJT habitat in the higher elevation portions of their range. These three factors alone, not to mention large-scale solar arrays, invasive species and the lack of the obligate pollinating moths, represent significant adverse cumulative impacts on the WJT.

One does not need to look into the foreseeable future to see if the WJT is, indeed, “threatened”. These threats are all occurring now.

Given these facts, as presented in the Department’s own Status Review, I urge the Commission to arrive at their own conclusion: that the Western Joshua Tree meets the definition of a Threatened species, as defined according to the California Endangered Species Act, and that they deserve the protections and regulatory safeguards that the CESA affords.

We cannot afford to “wait and see” as these iconic trees continue to decline over the next several decades until there are none. We/You—the Commission must act now to ensure that Joshua trees may continue to exist in their namesake National Park and on other protected lands in California.

Respectfully submitted,



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January 15, 2022

Subject: Peer Review of the California Department of Fish and Wildlife
Status Review of Western Joshua Tree

To Whom It May Concern:

Thank you for this opportunity to provide these peer review comments relative to the Draft Status Review of the Petition to List the Western Joshua Tree (WJT) as a Threatened species under the auspices of the California Endangered Species Act (CESA).

The primary purpose of the Status Review is to evaluate the appropriateness of listing the WJT as a Threatened species. According to the CESA, “An endangered species under CESA is one ‘which is in serious danger of becoming extinct throughout all, or a significant portion, of its range due to one or more causes, including loss of habitat, change in habitat, overexploitation, predation, competition, or disease’ (Fish & G. Code, § 2062). A threatened species under CESA is one ‘that, although not presently threatened with extinction, is likely to become an endangered species in the foreseeable future in the absence of special protection and management efforts required by [CESA]’ (*Id.*, § 2067). A species’ range for CESA purposes is the species’ California range (*Cal. Forestry Assn. v. Cal. Fish and Game Com.* (2007) 156 Cal.App.4th 1535, 1551). (Lines 3710-3719)

Given the extent of the range of WJT ($10,160 \text{ km}^2$ ($3,920 \text{ mi}^2$) to $13,880 \text{ km}^2$ ($5,360 \text{ mi}^2$), Lines 1382-1385), and the distribution of WJT within that range (3.1 million and 4.9 million adult western Joshua trees in California, Lines 1459-1465), the designation of WJT as an Endangered species, according to the definition under the CESA, is inappropriate at this time. The appropriateness of listing the WJT as a Threatened species, however, requires further definition of what is meant by the “foreseeable future”. As described in the Status Review, the “foreseeable future” with regard to the Threatened species definition is considered to be the 21st century, or by the year 2100.

The California Department of Fish and Wildlife (the Department) Draft Status Review provides a comprehensive and detailed description of the biology of the WJT (pages 5-29), its habitat and ecological parameters (pp. 30-37), its abundance/range and population trends (pp. 38-49), and endangerment factors (pp. 50-85).

These comments address each of these topics as they are presented in the Draft Status Review, with numeric line citations and quotations for reference, as necessary. Finally, I will review the Department’s conclusions and recommendations for the Petitioned Action, with my own suggestions as to the appropriateness or not of listing the WJT as a Threatened species under CESA.

Biology of the WJT

The Status Review provides an excellent synopsis of the taxonomy and biology of the WJT. For this peer review, I will limit my comments and analysis to the WJT—

Yucca brevifolia—as opposed to the Eastern Joshua Tree, *Yucca jaegeriana*. (Page 9, Lines 452-457).

The Petition states that the WJT warrants listing as a Threatened species throughout its range in California; but requests the Department consider listing the southern population (YUBR South) as “ecologically significant units”, as opposed to the northern extension of WJT (YUBR North)—generally the range of WJT from Inyo County, northward into Nevada and the Great Basin Floristic Province (Page 10-11, Lines 498-508 and Figure 3: Joshua Tree Range in California). As we will see in the discussion of endangerment factors, the levels of threat from land development, energy projects, wildfires and climate change are generally greater in the YUBR South range than the YUBR North range, thus warranting separate consideration of the appropriateness of listing under the CESA.

A key biological factor for the WJT is its obligate pollination mutualism with the yucca moth, *Tegeticula synthetica* (Page 19, Lines 719-730). The yucca moth and WJT are co-evolved to the degree that the WJT is dependent on the moth for sexual reproduction and the moth is dependent on the WJT for its own reproduction. Thus, it is not surprising that, “greater sexual reproduction tends to occur in areas with more *T. synthetica* moths (Harrower and Gilbert 2018). Within the vicinity of JTNP, *T. synthetica* moths were found at elevations ranging from 1,049 m (3,442 ft) to 2,076 m (6,811 ft), but not at the lowest elevation study site that had western Joshua trees at 1,004 m (3,294 ft) or the highest elevation study site with western Joshua trees at 2,212 m (7,257 ft) (Harrower and Gilbert 2018).” (Page 20, Lines 761-767)

Thus, the environmental limits of the yucca moth have a direct bearing on the sexual reproduction of the WJT, and the lower elevation limitations for the moth—most likely reflecting a high temperature threshold and/or low soil moisture tolerance—may indicate that these low elevation WJT populations are already no longer viable and will, with increasing temperatures resultant to climate change, become locally extinct.

Once fertilization of the WJT flowers has occurred and the moth larvae mature, they “fall to the ground below the tree, burrow into the ground, create a cocoon, and enter a period of suspended development called diapause (Pellmyr 2003). Yucca moth larvae are likely able to remain in diapause for several years before pupating into moths; the environmental or other cues that trigger this pupation are currently unknown (Riley 1892, Pellmyr 2003)” (Page 20, Lines 749-753).

Although asexual reproduction does occur in WJTs, particularly after fires and/or at higher elevations, sexual reproduction is essential for maintenance of genetic diversity of the species. Little is known about the life history or survival of yucca moths regarding their survival (or not) after fires, their environmental tolerances to extreme temperatures or moisture, or of their capabilities of locating host plants and dispersal in highly fragmented habitats, such as urbanized, low density WJT habitat in the YUBR South range. These potential endangerment factors relative to the *T. synthetica* moth are not addressed in the Status Review.

Other biological factors that are of critical importance in consideration of the endangerment of the WJT are summarized herein:

- Seed dispersal is very limited: average seed dispersal is ~30m (Lines 805-825)
- Seed germination requires periods of cooler, moist conditions for several years following mast seeding events. “After burial of seeds, several successive years of sufficiently wet and/or cool conditions are likely required to ensure that seeds germinate, and that seedlings reach a sufficiently large size (perhaps at least 25 cm) before the arrival of a period of hotter and/or drier conditions. This period of several successive years of sufficiently wet and/or

cool conditions must occur relatively soon after a mast seeding event, because western Joshua tree seeds do not remain viable in the soil for long periods of time.” (Lines 1077-1085)

- After germination takes place, seedlings require long periods of time, perhaps as much as 30-50 years, to reach reproductive maturity. “After a seedling has become established, it must survive a long period of time (perhaps 30-50+ years) to reach reproductive maturity. A minimum rate of recruitment is necessary to keep populations from declining (Wiegand et al. 2004).” (Lines 1083-1086)

These three biological factors all conspire to create a cumulative adverse impact on WJT health and viability in the face of the impacts of climate change: 1) that WJT seed dispersal is extremely limited and that dispersal to more northerly or higher elevation potential habitat will not keep pace with increasingly extreme high temperatures and drought; 2) the conditions of higher temperatures and drought at lower elevation WJT locations will adversely impact seed germination; and 3) the time from germination to reproductive maturity will be very slow, especially given the likelihood of increasingly severe heat and drought episodes, and the increasing frequency and severity of fires in the higher elevation populations.

Thus, just based on these biological requirements alone (not considering the impacts of land development in the middle elevation populations of the YUBR South range), we can expect the continuing loss of sexual reproduction in the lower and upper elevation populations of WJT; and an inability of WJT to adapt to these environmental extremes by dispersal to more northerly or higher elevation potential habitat.

WJT Habitat, Range and Ecological Parameters

The Status Review describes the pre-historic range of WJT: “Fossil evidence indicates that Joshua tree was more widespread during the late Pleistocene period (22,000 to 13,000 years before present) than it is today, with its range at that time extending south of its present range farther into southern California and Arizona, and likely also into northwestern Mexico (Rowlands 1978, Holmgren et al. 2010, Cole et al. 2011, Smith et al. 2011). The apparent reduction in Joshua tree range from the late Pleistocene period to modern times suggests the population trend of Joshua tree across its entire range has been in decline.” (Lines 1483-1488)

The contraction of WJT range in post-Pleistocene times has seen some expansion of WJTs to the north and into the Great Basin, and into higher elevation habitats on the southeastern Sierra Nevada slopes and the northern slopes of the Transverse Mountains and Little San Bernardino Mountains (Joshua Tree National Park). These transmutations of the species have taken place over several thousands of years.

The Status Review states in its Executive Summary that, “Predicted suitable habitat for western Joshua tree based on 20th century climate conditions will likely remain in some areas at the end of the 21st century as refugia, and newly appear to the north and in higher elevation areas, although western Joshua tree is unlikely to colonize areas with newly suitable climate conditions quickly.” (Lines 202-206)

Record high summer temperatures in recent years are already being measured in the lower elevations of the WJT range and increasing temperatures and overall reductions in precipitation will lessen recruitment of WJTs in those areas. For WJTs to “newly appear” to the north and in higher elevation areas implies that there would be some means of long-distance dispersal. During Pleistocene times, with much more temperate conditions than present, WJTs were dispersed over significant distances by giant ground sloths (Lines 857-878), whereas dispersal distances by desert packrats and other rodents is measured in meters and would be ineffective for WJTs to adjust to rapid climate change.

This Peer Review finds that the total range estimate of 10,160 km² (3,920 mi²) to 13,880 km² (5,360 mi²) for the WJT is reasonably accurate. Using the Department's WJT range map (Figure 4. Joshua Tree Absolute Cover), one can readily see that the higher WJT cover estimates (>1-5% and >5%) are all in the upper elevation range of the species, extending from the southeast Sierra Nevada slopes, the extreme western edge of the Mojave Desert west of Palmdale-Lancaster (where there is some supplemental precipitation through the Tehachapi Pass), and along the north aspect slopes of the San Gabriel, San Bernardino and Little San Bernardino Mountains.

Lower elevation areas of the WJT range are already exhibiting lower absolute cover and reduced seedling germination and recruitment. As the Status Review summarizes: Declines due to reduced seedling recruitment will likely be most severe in areas of western Joshua tree's range that are already near the thermal and water stress tolerance limits for recruitment, such as at hotter, low-elevation areas. (Lines 1970-1972)

The compounding endangerment factors of climate change are described further by the Department: "The climatic conditions across western Joshua tree's range have already changed and will continue to change as a result of ongoing global carbon emissions. The Department expects that the direct effects of climate change (e.g., increased temperatures and decreased total water availability locally) will likely contribute to a decline in populations of [WJT] within California through the end of the 21st century... The primary reasons for the decline of populations of [WJT] within California may be the incremental contribution of climate change to high intensity and longer duration droughts, coupled with extreme high temperatures during summer months, which may have direct physiological effects on [WJT] plants. These effects of climate change will likely reduce [WJT] seedling recruitment, and to a lesser extent also increase adult [WJT] mortality, leading to population declines as recruitment does not keep pace with mortality. Climate change may also contribute to the decline of populations of [WJT] via other more indirect mechanisms, including increased impacts from small mammals during drought, reduced growth due to lack of low winter temperatures, increases in fire activity, or effects on pollinating moths..." (Lines 1914-1930).

Furthermore, "There may be a time delay between the time when an area becomes no longer suitable for a species (crossing an extinction threshold) and when that species is no longer present, (Tilman et al. 1994, Kuussaari et al. 2009, van Mantgem et al. 2009, Svenning and Sandel 2013, Figueiredo et al. 2019). Extinction processes often occur with a time delay and populations living close to their extinction threshold might survive for long periods of time despite local extinction being inevitable (Hanski and Ovaskainen 2002, Lindborg and Eriksson 2004, Helm et al. 2006, Vellend et al. 2006, Malanson 2008, Cronk 2016). Because western Joshua tree is a long-lived species, adults could persist for decades or longer in areas that are no longer suitable for recruitment, or recruitment may continue, but at rates that are ultimately insufficient to maintain the species. Although these areas may appear occupied, the presence of western Joshua tree may merely represent a delayed local extinction. (Lines 2018-2029)

Thus, when one re-examines the range of YUBR South as illustrated in Figure 4, one can see that fully half of the total YUBR South distribution may already be functionally extinct—that is, non-reproductive at rates that can sustain the population in those areas in the "foreseeable future" (the 21st century).

Endangerment Factors

The Status Review examines three primary factors affecting the survival and reproduction of the WJT: climate change, [land] development and other human activities, and wildfire. Other factors, including invasive plants, herbivory and predation, and human use and vandalism are not considered to be significant endangerment factors unto themselves and are not discussed further in this Peer Review.

Climate—

The potential impacts of climate change have been described in the previous section of this peer review. Several climate models are presented in the Status Review: Thompson et al. 1998, Shafer et al. 2001, Dole et al. 2003, Cole et al. 2011, Barrows and Murphy-Mariscal 2012, and Sweet et al. 2019. The first four models evaluate both Eastern and Western Joshua Tree species; and the last two are focused on WJT population models for JTNP and vicinity.

The Status Review summarizes the models' findings: "The species distribution modeling efforts that have been conducted for Joshua tree suggest that climate change could cause substantial reductions in areas with 20th century suitable climate conditions for the species at the southern parts of western Joshua tree's range, including within JTNP. These species distribution modeling efforts also suggest that substantial additional areas of 20th century suitable climate conditions may become available for western Joshua tree to the north, particularly in Nevada (outside of the scope of CESA) but also in some parts of eastern California, although **the species is unlikely to naturally colonize these areas in the foreseeable future.**" [Bold highlight added by the Peer Reviewer]

The models all indicate a contraction of WJT range from lower elevation slopes, where extreme summer high temperatures and increasing drought will cause those areas to become locally extinct, toward higher elevation or northerly areas characterized by cooler temperatures and more precipitation. The Status Review is correct, however, in qualifying that "the species is unlikely to naturally colonize these areas in the foreseeable future," because of its inability to disperse such long distances over inhospitable terrain, given the species' short dispersal range (~30m, op.cit. under Biology of the WJT).

Land Development—

Land development in the form of clearing the land for agriculture, housing and urban development, or energy projects represents a direct and permanent loss of WJT habitat. Most land development in the Mojave Desert region occurs on private land in the YUBR South range.

The Status Review mentions large-scale clearing of land in the western Mojave Desert portion of the range for dry farming and agriculture during the early 1900s and how these areas have shown little or no WJT recruitment since those times. Much of that area has since been developed for housing and urban development in the cities of Palmdale and Lancaster.

The Status Review reports that, "Much of the recent western Joshua tree habitat modification and destruction has been the result of ongoing urban development, typically on private property within the general vicinity of existing developed areas. The USFWS (2019) reported that approximately 50% of the southern part of western Joshua tree's range (YUBR South) is on private property, 2% of the northern part of western Joshua tree's range (YUBR North) is on private property, with the remainder predominately on federal land. WEST Inc. (2021b) found a higher percentage of western Joshua tree's range on private property than the USFWS did, with approximately 65% of the southern range on private property..." (Lines 2562-2570)

An unpublished recent study conducted by a Geographic Information Systems (GIS) research group under my direction at the University of Redlands found that 420 mi²/677 km² of WJT habitat within the cities of Palmdale, Lancaster, Yucca Valley, Joshua Tree, Twentynine Palms, Victorville, Hesperia, and Apple Valley were developed within those jurisdictions between 1984 to 2021 (Krantz et al. 2021). This analysis examined decadal aerial photo imagery, identifying developed areas within those jurisdictions, but it did not include isolated blocks of open space that may represent occupied WJT habitat. In fact, the remaining undeveloped blocks within these cities are

so highly fragmented that they likely no longer represent ecologically viable habitat. Given the extremely short distance dispersibility of WJT seeds and isolation from potential yucca moth pollinators, these remaining patches of WJT habitat should be considered ecologically unviable and essentially extirpated.

Within the foreseeable future (the year 2100), if not already, the undeveloped areas of these incorporated cities should be considered functionally extinct. Most of the smaller fragments of extant habitat are already ecologically unviable and would, therefore, meet the definition of functionally extinct, as described in the previous section of this peer review.

City	Area (mi ²)
Palmdale	106.3
Lancaster	94.54
Victorville	74.01
Hesperia	72.78
Adelanto	52.88
Apple Valley	77.08
Joshua Tree	37.04
Yucca Valley	39.83
Barstow	41.34
Twenty-nine Palms	58.76
	654.56 mi ²

If one considers the incorporated cities within the YUBR South range as developed habitat within the foreseeable future, then a total habitat loss of 654.56 mi² should be considered extirpated and functionally extinct.

Fifteen renewable energy projects were granted §2084 take exemptions during the hearings to establish the WJT as a candidate species for listing under the CESA. According to an analysis done by the USFWS using U.S. Environmental Protection Agency Integrated Climate and Land Use Scenarios projections, between 22% and 42% of the habitat within the southern part of western Joshua tree's range may be lost by the year 2095 due to urban growth and renewable energy development. (Lines 2641-2645)

Wildfire

Wildfire, although a defining component in many of California's ecosystems, is a relatively rare phenomenon in the Mojave Desert, but fire frequency and intensity has increased dramatically in recent decades, especially in the period from 2001-2020, as illustrated in *Figure 9: Fires within the California Range of Western Joshua Tree, 1900-2020* (CALFIRE 2021) of the Status Review.

Within the WJT range, "Fire is unevenly distributed in the Mojave Desert, and fire occurrence tends to align with distinct precipitation regime boundaries, with most large and recurring fires occurring in areas that have a relatively high amount of precipitation in summer (Tagestad et al. 2016). Fuels tend to be more available, and fires tend to be more frequent and severe at higher-elevation areas of the Mojave Desert, and the availability of fuels and frequency of fires is somewhat lower at middle elevation areas, and still lower at the low elevation areas of the Mojave Desert (Brooks et al. 2018). (Lines 2683-2690)

The size, intensity and frequency of fires in the YUBR South range are the result of higher fuel loads in the higher elevation portions of the species' range and increasing

drought and higher summer temperatures—characteristics of climate change. The GIS study completed by Krantz et al. (unpublished, 2021), using the same CALFIRE database as cited in Figure 9, above, estimated that between 1980-2019 a total area of 950km² of WJT habitat was burned within the YUBR South range, representing approximately 8% of total WJT habitat, but as much as 12.9% of YUBR South distribution.

Wildfire impacts on YUBR habitat are severe. As cited in the Status Review, “Western Joshua tree populations are very slow to recover from fire. Minnich (1995) observed a 47 year chronological sequence of 13 burned Joshua tree woodland sites within JTNP that were similar, but that had burned at different times in the past. Minnich (1995) found that 64% to 95% of western Joshua tree stems were fatally damaged in all but one of the sample sites, and western Joshua tree cover and density remained low in burned sites compared with unburned sites, even 47 years after burning.”

Smaller WJT plants (<0.5m) are almost entirely killed by fire, but even taller, mature trees are largely killed above ground. These may sprout vegetatively after fires, but these sprouts may take 30-50 years before reaching sexual maturity and producing flowers.

The Department summarizes the impacts of wildfire on the WJT as follows: “Wildfire is a substantial threat to western Joshua tree and invasive plants contribute to that threat, but wildfire does not affect the entire range of the species evenly, does not necessarily burn through habitat in a uniform, high-intensity way, and does not typically result in the complete elimination of western Joshua tree from burned areas. For these reasons, wildfire is likely to reduce the abundance of the species, but it is unlikely to result in a serious danger of elimination of the species throughout a significant portion of its range. Nevertheless, because western Joshua tree recruitment from seed is rare, and because the species takes a long time to reestablish in burned areas, wildfire causes long-lasting negative effects in burned areas. The Department expects that the impacts from continuing and increasing wildfire activity in the Mojave Desert and surrounding areas will cause ongoing gradual reductions in the size of at-risk populations of western Joshua tree within California, but the range of the species is unlikely to be affected by wildfire in the foreseeable future, because western Joshua tree is unlikely to be completely eliminated from affected areas due to its high abundance and widespread distribution.” (Lines 2893-2907)

This conclusion fails to account for several factors. With increasing fire frequency and intensity, vegetative sprouts of WJTs are largely eliminated from these areas if the subsequent fire occurs before the sprouts are more than 2-2.5m high—the height at which Southern WJTs first flower (Rowlands, 1978). Fires eliminate seed stock in the soils and remove potential nursery plants, further reducing the potential for flowering, seed production and seed germination for the “foreseeable future”—the end of this century. Finally, studies cited in the Status Review indicate that the yucca moth, upon which the WJT is dependent for pollination, is already rare at these higher elevations of the WJT range (Harrower and Gilbert 2018). With the elimination of flowering YUBR plants for 50+ years (before vegetative sprouts will flower again), these areas are essentially lost for their requisite pollinators.

Conclusions Regarding Listing the WJT as a Threatened or Endangered Species

It is clear that the Western Joshua Tree does not meet the definition of an Endangered species in accordance with the CESA: a species “which is in serious danger of becoming extinct throughout all, or a significant portion, of its range.” The question before the Department and the focus of this Status Review is whether the WJT meets the definition of a Threatened species, a species “that, although not presently threatened with extinction, is likely to become an endangered species in the foreseeable future in the absence of special protection and management efforts required by [the CESA].” In this Status Review, the foreseeable future is considered to be the 21st century, or through the year 2100.

Furthermore, the focus of this Peer Review is on whether the southern extent of the populations of WJT (YUBR South) should be listed as Threatened “ecologically significant units” unto themselves. The conclusion of this Peer Review is that such a designation is appropriate because the YUBR South populations are subject to much greater threats than the northern WJT populations (YUBR North). The facts and research presented in this Status Review demonstrate that the potential for the YUBR South populations to become Endangered over a significant portion of their range (the YUBR South range) within the foreseeable future is very real.

The primary threats to the YUBR South populations of WJT are three-fold:

- Climate change
- Urbanization and land development
- and Wildfires

Each of these endangerment factors will be discussed briefly and summarized below.

Climate Change

The impacts of climate change are already manifest in the YUBR South populations, with higher summer temperatures and more extreme drought, particularly in the lower elevations of the YUBR South range. As stated in this Status Review, “climate change could cause substantial reductions in areas with 20th century suitable climate conditions for the species at the southern parts of western Joshua tree’s range, including within JTNP.”

As the climate models cited in this Review have found, the impacts of increasing heat and drought will be most severe at the lower elevations of the YUBR South range.

The Status Review goes on to state that, “These species distribution modeling efforts also suggest that substantial additional areas of 20th century suitable climate conditions may become available for western Joshua tree to the north, particularly in Nevada (outside of the scope of CESA) but also in some parts of eastern California, although the species is unlikely to naturally colonize these areas in the foreseeable future.”

The Department is correct in noting that as the climate warms and low elevation areas of the WJT become uninhabitable for the species, other areas to the north and at higher elevations may develop suitable climate conditions; but the Department is also correct in stating that the species is unlikely to naturally colonize these areas in the foreseeable future (by the Year 2021) due to its very limited dispersibility (~30m).

It will be virtually impossible for WJT in the southern populations to disperse over these relatively few decades to the northern YUBR populations to adjust to climate change. As far as Southern YUBR plants are concerned, dispersal to newly suitable habitat in the YUBR North range is not possible. They will have to disperse/migrate to the higher elevation, cooler, moister habitats of the slopes along the southern edge of the YUBR South range, which we will see below, is also impossible.

In fact, the lower elevation populations of southern WJT are already experiencing very low reproduction rates and those individuals are not maturing to achieve mature flowering plant status, as illustrated in Figure 4 of the Status Review. Furthermore, there is evidence that the obligate pollinator, *T. synthetica*, may already be declining or absent from the lower elevation WJT populations, implying that these areas cannot sexually reproduce (Harrower and Gilbert 2018) and, therefore, cannot produce viable seeds for dispersal. The lower elevation portions of YUBR South should already be considered ecologically and functionally extinct.

To visualize the extent of the impact of climate change on the YUBR South metapopulation, the entire area shown as yellow on Figure 4 will be functionally extinct within the foreseeable future. Yes, there will be islands of refugia in the isolated mountains north of Barstow and northeast of Lancaster, but these islands will be reproductively and ecologically isolated to the extent that they are biologically doomed if current climate trends continue, as the climate models cited in the Status Review all predict.

Urbanization and Land Development

As described earlier in this Peer Review, urbanization and land development in the desert cities of the southern Mojave Desert represent an enormous and permanent conversion of WJT habitat. Development and fragmentation within the incorporated city limits alone represent more than 650 square miles of habitat loss. This does not include the clearing and destruction of the 15 large-scale renewable energy projects that were granted §2084 take exemptions during the hearings to establish the WJT as a candidate species for listing under the CESA, nor does it include the expansive developments of rural “ranchettes” and other associated developments in the unincorporated communities of Phelan, Oak Hills, Baldy Mesa, Lucerne Valley, and Pioneertown, to name just a few. (See attached Image of the Unincorporated Communities)

Most of this development, from the Palmdale-Lancaster area in the western portion of the YUBR South range, to the cities of Yucca Valley and Joshua Tree at the eastern limit of the YUBR South range, extends across the middle elevations of the southern WJT habitat. Remaining fragments of occupied habitat within these city limits are, once again, functionally extinct. That is, extant WJTs on these remaining patches are now totally isolated, unable to disperse to higher ground in the face of warming temperatures and increasing drought. This isolation is compounded by the fact that they require the presence of yucca moths for pollination and production of viable seeds; and even if pollination is successful, the dispersal of seeds across the fragmented urban landscapes becomes increasingly unlikely, if not impossible.

Furthermore, the development of the wide swath of the middle elevations across the southern flank of YUBR habitat effectively isolates the entire lower elevation populations from any chance of dispersal across the urban barrier to reach the cooler, moister suitable habitats in the face of climate change. This compounds the effective isolation of the lower elevation populations, reinforcing their functional extinction.

Wildfires

Finally, we have the fact of increasing frequency, size and severity of wildfires in the southern WJT range. As noted in the Status Review (see Figures 9 and 10), the area burned by wildfires has more than doubled in the last three decades in comparison with the previous 90 years. Most of those fires and the largest of them have occurred in the higher elevations of the YUBR South range.

For example, the Sawtooth Fire Complex near the community of Pioneertown (readily visible on Figure 9 in the southeast portion of the range), consumed 61,700 acres of high quality, high density WJT woodland habitat in 2006. Now, 15 years later, the area is still nearly devoid of WJT plants, with no mature Joshua trees in the burn area and very few vegetative sprouts. The area is essentially “dead” for many generations to come, with no flowering WJT plants. The lack of mature, flowering Joshua trees equates

to no yucca moths. The absence of the flowering host plant will eliminate the yucca moths from the area for many moth generations, certainly for the “foreseeable future” through the Year 2100.

The WJT and yucca moth are obligate co-dependent species. This represents a significant and cumulative adverse impact, with very serious implications for WJT in wildfire areas. It means that these areas, even if they recover by vegetative reproduction from the fire, will remain without their obligate pollinators for many decades or even beyond 2100.

The fact that these wildfires are almost entirely in the higher elevations in the southernmost extent of the YUBR South range effectively removes the climate refugia that lower elevation populations will need, if they are capable of dispersal to these cooler, more hospitable habitats at all.

Final Peer Review Recommendation

Thus, we find that the Southern WJT populations are faced with a triple cumulative threat: their lowermost populations are already functionally extinct due to climate change; even if they could disperse toward higher, more equable climate, they are blocked by sprawling development across their middle elevations; and finally, the remaining high ground along the south edge of the YUBR range is being consumed by wildfire and will be biologically non-functional for the foreseeable future and beyond.

Together, these three impacts represent significant adverse cumulative impacts to the YUBR South populations *throughout their range*. Referring back to the definition of an Endangered species: one “which is in serious danger of becoming extinct throughout all, or a significant portion, of its range;” I find that the data and studies presented in this Status Review do, indeed, support a finding that the YUBR South population of WJT meets the definition of a Threatened species: one that, “although not presently threatened with extinction, is likely to become an endangered species in the foreseeable future in the absence of special protection and management efforts required by [the CESA].”

Management Recommendations and Recovery Measures

If the Department finds that the WJT does not warrant protection as a Threatened species under the CESA, then other regulatory and recovery measures shall be necessary to ensure that the species does not become a Threatened species.

The Department lists a range of management recommendations and recovery measures (Lines 4056-4088). A few of these measures are practical and may be implemented, while many are vague, impractical and unenforceable. I will briefly review the recommendations below.

- 1) Continue efforts to drastically reduce greenhouse gas emissions. Of course!
- 2) Complete a western Joshua tree conservation plan in collaboration with partners and stakeholders. The WJT Conservation Plan should include detailed protocols for environmental assessment and mitigation of proposed projects that have the potential to impact WJTs.
- 3) Preserve western Joshua tree habitat in areas with high recruitment and areas projected to be climate refugia. Dedicate State funds toward acquisition and protection of otherwise unprotected high-value WJT habitat.
- 4) Minimize wildfire risk to western Joshua tree woodlands, particularly following one or more years of high precipitation, and particularly in areas with high recruitment and areas projected to be climate refugia.

Would this mean weed-whacking non-native flash fuels over hundreds of square miles?
Impractical.

5) Manage fires aggressively to protect Joshua tree woodlands, particularly in areas with high recruitment and areas projected to be climate refugia. Not practical.

6) Implement ways to disincentivize destruction of western Joshua tree habitat, particularly in areas with high recruitment and areas projected to be climate refugia. What sort of “disincentives” are contemplated here? Not practical.

7) Implement state and/or local laws and regulations that limit unmitigated impacts to high quality western Joshua tree habitat. Not practical unless accompanied by enforceable, regulatory measures.

In this circumstance, it is my recommendation that the Department sanction the WJT in its YUBR South distribution as a Regulated species, like regulated game or fish animals.

8) Continue scientific investigations into the biology, ecology and genetics of western Joshua tree and the species and habitats upon which it depends:

- o Collect and analyze range-wide demographic information to detect baseline population trends and identify populations that do not appear to be recruiting new individuals at sustainable levels.
- o Implement long-term range-wide direct population monitoring and vegetation monitoring with emphasis on leading and trailing edges and highest and lowest elevations of the species’ range.
- o Produce and improve upon range-wide species distribution models for western Joshua tree.
- o Produce range-wide species distribution models for western Joshua tree’s obligate pollinating moth.
- o Investigate the significance of multi-year and multi-decade climate variability patterns for western Joshua tree recruitment.
- o Investigate ways to control the spread and abundance of invasive plant species to reduce wildfire risk.
- o Investigate the feasibility, practicality, and risks of implementing assisted migration and translocation.

Of these last measures, all are necessary to provide basic baseline monitoring information for the WJT. Of particular importance would be to promote further investigations and biological research on the obligate pollinating moth, *Tegeticula synthetica*. The Status Review presents some basic information about the life history of the moths, but certain information pertinent to this Petition is lacking, such as: what are the temperature and moisture thresholds for the species? There is some indication that the moths are rare or absent at the lower and upper elevations of WJT. What are the limiting factors that determine its range? These are **obligate**, co-dependent species. Therefore, the limiting environmental factors of one have direct consequences on the distribution of the other.

One of the more practical measures, not mentioned above, would be to require consideration of projects within the YUBR South range to undertake environmental impact assessments in accordance with the CEQA guidelines. The Status Review describes this alternative (Lines 4007-4019), but, without formal listing, there would be no requirement that projects analyze potential impacts to WJT.

If, however, the State designated the Southern WJT as a Regulated species, similar to other game and fish animals (§2116 *et seq.* of the Fish and Game Code), then CEQA review or at least regulatory review would be required, and permits would be necessary for removal of WJT plants on impacted properties. By this means, projects that have the potential to adversely impact WJTs would have to consider avoidance of WJTs to the extent possible and mitigation of impacts to WJTs in the case that Joshua trees cannot be avoided.

Regarding mitigation for removal of WJT, the trees may be successfully transplanted. San Bernardino County enacted a Joshua tree policy in the late 1980s that required developers to avoid the trees if possible, translocate them or make them available for translocation if necessary. During this time and through the 1990s, I worked with a landscape company, *NativeScapes*, transplanting Joshua trees using a 24-inch and

36-inch hydraulic tree spade. Joshua trees have a fibrous root system, like palm trees, and they can be excavated and placed in 36-inch or 48-inch boxes for re-location to protected areas on- or off-site. Trees as tall as 10-12 feet with moderate branching can be transplanted.

Once the trees are installed, larger trees must be tethered to stabilize the weight of the tree; and transplants must receive additional irrigation maintenance through the first two summer seasons until the fibrous root system is reestablished.

For this practice to be effective, it is essential that the State designate the WJT as a regulated species. Otherwise, if left to the individual county and city municipalities, the southern WJT would have only inconsistent standards for environmental review and mitigation. Standardized environmental assessment and mitigation measures may be included in the WJT Conservation Plan recommendations, described in #2 above.

The WJT Conservation Plan may also identify conserved areas for translocation of Joshua trees in anticipation of climate change, such as the Pioneertown Preserve. The Pioneertown Preserve is a 25,500-acre natural preserve managed by The Wildlands Conservancy. The area was burned during the Sawtooth Complex Fire in 2006 and native WJT woodland habitat has been very slow to recover. Translocation from lower elevation sites in the cities of Yucca Valley and Joshua Tree to the Pioneertown Preserve would facilitate WJT recovery from the fire, as well as help with climate adaptation by moving plants to higher elevations. Such translocation sites would require long-term management for fire and fuel modification, non-native grass and fuels management around the base of the trees, and irrigation maintenance until such trees are re-established.

Other potential “climate refugia” may be identified in the Conservation Plan on State, Federal or private lands across the WJT range.

This concludes my Peer Review comments on the Status Review of the Petition to List the Western Joshua Tree. Thank you again for the opportunity to provide these comments. If the Department has any further questions in these regards, please do not hesitate to reach out to me at the numbers/email below.

Sincerely,



Dr. Timothy Krantz
Professor and Chair, Environmental Studies Program
University of Redlands
1200 E. Colton Avenue, Lewis Hall 121
Redlands, CA 92374



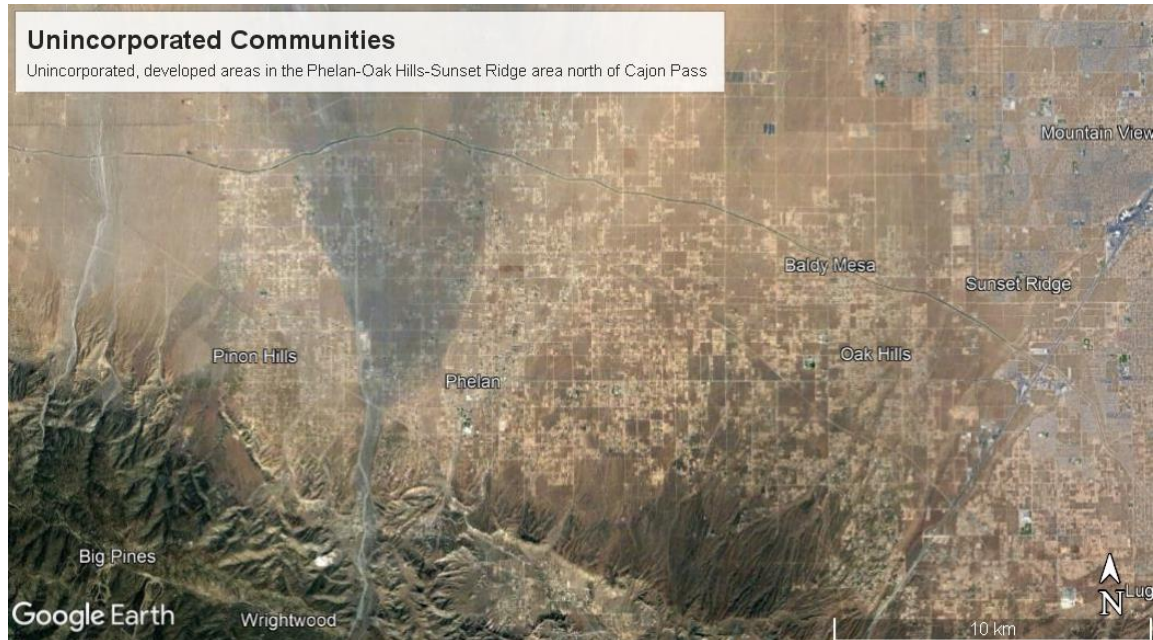


Figure: Unincorporated Communities in the Phelan-Baldy Mesa Area.

Note the extensive land clearance for small ranches and rural residential development. These communities are entirely within high density WJT woodland habitat. Estimated WJT habitat loss just within this image is approximately 300 km². The even more densely developed cities of Hesperia and Adelanto are immediately east of the image.



Kenneth S. Norris Center for Natural History
1156 High Street
Santa Cruz, CA 95064

California Fish and Game Commission
Re: Western Joshua tree status review

June 2, 2022

Dear President Murray and Commissioners:

Thank you for granting Joshua trees interim protection in 2020. With the boom in development and solar industry, this was a key decision towards securing Joshua tree populations. The continued protection of Joshua trees is very important if we want to maintain population viability moving forwards.

I am a Joshua tree climate change ecologist and I disagree with the recent status review by the CDFW which concluded that protections are not warranted. I agree with one of the reviewers who stated that the review reads as an argument for not listing the species as threatened as opposed to an objective analysis of the science. There is an underestimation of risk throughout the report as well as missed opportunities to answer questions posed by the department with scientific data that is already available.

Firstly, numerous peer-reviewed climate modeling studies of Joshua tree distribution were largely dismissed because they were species distribution models, which in fact do provide very useful predictive ecological data (this was a similar issue in the federal status review which denied Joshua trees listing, and was actually overturned). The report continually states that there is no information on how climate change will actually impact Joshua trees or that populations may cease to be reproducing in certain locations. The status review also claimed a lack of demographic and climate data, however, they cited my peer-reviewed Joshua tree climate demography research over 20 times in the report. This is confusing as my work directly demonstrates these impacts.

In my research¹ I found no pollinators or seedling recruitment at either the hottest/driest location or the coolest locations along a climate gradient at the southern-most range for Joshua trees. This suggests that the southern range edge of Joshua trees is contracting but also not shifting upwards into higher elevation. I also found high tree mortality at both locations. Perhaps climate change will make those higher locations more hospitable to future Joshua trees, but they are also locations of high fire threat due to invasive grass species. Regardless, seedlings and their obligate pollinators are not currently moving upslope into those areas. In my recent research on Joshua trees and their soil mycorrhizal symbionts², I found that fungal species change along an elevation gradient and that these different species interactions may have significant consequences for Joshua drought tolerance and seedling recruitment at range edges. These studies document important components of Joshua tree ecology that are compromised under climate conditions and will only multiply with largescale tree removal on private lands across CA.

The status review ignores these and other recent data on climate related impacts to Joshua tree habitat, instead utilizing recruitment data from the 1990's as a basis to assert that Joshua tree populations are healthy and stable, assertions that conflict with more recent data from myself and others that show

¹ Harrower, J. and G. S. Gilbert. 2018. Context-dependent mutualisms in the Joshua tree-yucca moth system shift along a climate gradient. *Ecosphere* 9(9): e02439. 10.1002/ecs2.2439.

² Harrower, J.T. and G.S. Gilbert. 2021. Parasitism to mutualism continuum for Joshua trees inoculated with different communities of arbuscular mycorrhizal fungi from a desert elevation gradient. *PLOS ONE*.
<https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0256068>

that the majority of Joshua trees at their southernmost ranges (hottest and driest) are either not recruiting seedlings or have reduced recruitment.

Overall, throughout the report, there is a pattern of refuting or misunderstanding solid data. The requirement is for the state to use the best available science to base its recommendations for listing as opposed to putting off protections for future decades until more data can be collected. The profound understatement of the impact of climate threats as summarized by the department is concerning and is not representative of actual climate science (for example the recent findings in the IPCC report). There is also no discussion of the combined and additive impacts of the many threats that are listed. The general dismissal argument that Joshua trees are widespread and abundant and thus not warranting protection is a severe short sight by the department and a meaningless argument.

I was born in the Mojave and have spent over a decade researching Joshua tree habitat within or near the national park. My family lives locally, and I have experienced firsthand how bad the local management of Joshua trees is by the town of Yucca Valley and San Bernardino County. With the booming Airbnb economy (Joshua tree is one of the top 2 fastest growing markets in CA) we need to effectively manage Joshua tree habitat given the increased interest in development. While I am happy to see that San Bernardino passed an ordinance in March 2022 which raises the fines for illegally removing Joshua trees, I suspect that this has more to do with an attempt at demonstrating that the county is now attempting to protect Joshua trees and that they are hoping that this will convince the commission that there is no need for continued state protection. It is worth noting that this doesn't change the exemptions for permit issuance, and regardless, enforcement by the county has always been very low. Additionally, this also does nothing to address the severe climate change and fire threats to Joshua tree populations.

Maintaining healthy Joshua tree populations is also an important aspect of the tourism economy. I have given numerous public talks on Joshua tree ecology around the country to thousands of people, many whom have visited or plan to visit the Mojave Desert. The typical stated reason that people want to visit the area is their desire to see actual living Joshua trees.

Joshua trees would be the first species protected in California due to climate change. It is important that California continues in its legacy of true climate leadership - we don't have the time to wait for more data to protect these species. As a majority of the Joshua tree habitat in the southern half of its range is on private property in CA, it is vulnerable to threats by largescale development projects. I hope that you follow the best available science with your decision and choose to prioritize the protection of this iconic species for our future generations.

Sincerely,



Jennifer Harrower, PhD
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June 2, 2022

Samantha Murray, President
California Fish and Game Commission
1416 Ninth Street, Suite 1320
Sacramento, CA 95814

Submitted via email to fgc@fgc.ca.gov

RE: Agenda Item 5: Petition to list Western Joshua Trees as threatened under the California Endangered Species Act – Support “Threatened” Listing

Dear President Murray and Commissioners:

Thank you for the opportunity to provide comments on your upcoming decision as to whether listing western Joshua trees as “threatened” under the California Endangered Species Act (CESA) is warranted (Fish & G. Code, § 2075.5).

The undersigned organizations, which represent hundreds of thousands of people throughout California, strongly support continued state protection of the western Joshua tree. We urge the Commission to adopt a

finding that listing of the species *is* warranted and officially add the species to the list of Threatened Species. Doing so would be a demonstration of true climate leadership, as the western Joshua tree would become the first species protected under CESA primarily due to the threat of climate change.

Our state's western Joshua trees are being profoundly impacted by human activity, most directly from climate change and habitat loss, but also from wildfire, drought and invasive species, each of which are exacerbated by climate change. Protections under CESA would greatly help protect western Joshua trees from direct habitat loss as well as foster recovery actions by the California Department of Fish and Wildlife (DFW) and other state and local agencies. These protections are necessary to ensure that this iconic species continues to inhabit California and inspire future generations.

While much of the western Joshua tree's range is on public lands, over 40% of its California range is on private land, of which only a tiny fraction is protected from development. Outside of Joshua Tree National Park, areas of federal land that are home to the species are subject to poorly regulated activities including off-road vehicle use, cattle grazing, power and pipeline rights-of-way and large-scale energy projects that consume or degrade habitat.

The Joshua tree's ability to colonize new habitat at higher elevations or latitudes is extremely limited and no such range expansion is yet occurring, even as the lower elevation and southern edge of its range is already contracting in the face of a warming climate. The convergence of factors necessary for recruitment results in successful establishment of new Joshua tree seedlings only a few times a century. Such recruitment has already largely stopped at the drier, lower limits of the species' range.

In this context, climate change represents an existential threat to western Joshua trees. The western Joshua tree in California will lose upwards of 90% of its range under likely climate scenarios. There is no safe refuge for western Joshua trees, as the higher elevation areas in which Joshua trees are projected to best be able to survive increasing temperatures and drying conditions are at great risk of fire due to the prevalence of invasive non-native grasses. Prolonged droughts, which are projected to occur with greater frequency and intensity over the coming decades, will not only preclude recruitment across ever-greater areas of the species' range, but will lead to higher adult mortality, either directly due to temperature and moisture stress or indirectly due to increased herbivory from hungry rodents lacking alternative forage. Further, it is uncertain whether the western Joshua tree's pollinating moth will be able to keep pace with a changing climate. Absent the pollinating moth, Joshua trees will not be able to produce seeds, meaning there will be no juvenile trees to replace older trees as they die off.

Additionally, the western Joshua tree is further threatened by direct habitat loss. Development has already consumed vast swaths of habitat in the range of the western Joshua tree. Over the coming decades, over a million additional acres are projected to be destroyed or degraded from development. This large-scale loss or severe degradation of habitat would be of conservation concern for the species even ignoring the threats posed by climate change. Taking climate change into account, such loss of habitat and the genetic resiliency and connectivity it provides will further push the species towards extinction in California.

Without rapid and substantial reductions in greenhouse gas emissions and protection of habitat, the western Joshua tree will likely be extirpated from most if not all of its range in California by the end of the century. It therefore clearly meets the definition of a "threatened species" under state law and must be protected as such.

While the threats facing the western Joshua tree in the coming decades are dire, they are not insurmountable. If the species and its habitat are protected early, and with active management to enhance recruitment and survival, and potentially dispersal, the western Joshua tree has a realistic chance of persisting in the wild over the coming decades. CESA listing, and the consequent development and implementation of a recovery plan and local or regional Natural Communities Conservation Plans (NCCPs), would help ensure the survival of this iconic species of the Mojave.

We recognize that recognizing a species as warranting protection under CESA primarily due to the threat of climate change is something that DFW and the Commission have never done. We also recognize that DFW recommended against listing the western Joshua tree, and voting to protect a species notwithstanding an adverse recommendation of DFW is not a decision to be made lightly. Nevertheless, given that four of the five independent peer reviews (the same four peer-reviewers who have actually studied the species) disagreed either with DFW's recommendation or with the rationale behind it, we believe that the overwhelming weight of the science still supports listing.

Rather than becoming yet another tragic symbol of our political leaders' failure to adequately respond to the climate crisis, protection under CESA would allow the western Joshua tree to serve as a symbol of California's climate leadership. We look forward to you demonstrating such leadership and making the legally and scientifically necessary finding that listing the western Joshua tree as a threatened species *is* warranted.

Sincerely,

Ileene Anderson
Senior Scientist/Public Lands Deserts Director
Center for Biological Diversity

Michael Madrigal (Cahuilla)
President
Native American Land Conservancy

Nicolas Jensen, PhD
Conservation Program Director
California Native Plants Society

Maria Jesus
Conservation Chair
Bristlecone Chapter
California Native Plants Society

Lucinda A. McDade, Ph.D.
Executive Director
California Botanic Garden

Erin Woolley
Policy Advocate
Sierra Club California

Janessa Goldbeck
Chief Executive Officer
Vet Voice Foundation

Jocelyn Silverlight
Executive Director
Friends of Big Morongo Canyon Preserve

Mason Voehl
Executive Director
Amargosa Conservancy

Sam Easley
Executive Director
Transition Habitat Conservancy

Jora Fogg
Policy Director
Friends of The Inyo

Steve Bardwell
President
Morongo Basin Conservation Association

Chris Clarke
Ruth Hammett Associate Director
California Desert Program
National Parks Conservation Association

Kelly Herbinson
Cody Hanford
Joint Executive Directors
Mojave Desert Land Trust

Jack Thompson
Desert Regional Director
The Wildlands Conservancy

Drew Feldmann
Conservation Chair
San Bernardino Valley Audubon Society

Travis Longcore
President
Los Angeles Audubon Society

Robert Parker
President
Kerncrest Audubon

Catherine Rich
Executive Officer
The Urban Wildlands Group



CALIFORNIA
NATIVE PLANT SOCIETY

2707 K Street, Suite 1, Sacramento, CA 95816-5130 (916) 447.2677 www.cnps.org

Protecting
California's native
flora since 1965

June 2, 2022

Samantha Murray, President
California Fish and Game Commission
1416 Ninth Street, Suite 1320
Sacramento, CA 95814

Submitted via email to: fgc@fgc.ca.gov
CC: Jennifer.Greaves@FGC.ca.gov

Re: Agenda Item 5: Petition to List Western Joshua Trees as Threatened Under the California Endangered Species Act - Support "Threatened" Listing

Dear President Murray and Commissioners:

Thank you for the opportunity to comment on the upcoming decision regarding listing the western Joshua tree (*Yucca brevifolia*) as "threatened" under the California Endangered Species Act (CESA). The following comments are submitted on behalf of the California Native Plant Society (CNPS), a non-profit environmental organization with over 11,000 members in 35 Chapters across California and Baja California, Mexico. CNPS's mission is to protect California's native plant heritage and preserve it for future generations through the application of science, research, education, and conservation. We work closely with decision-makers, scientists, and local planners to advocate for well-informed policies, regulations, and land management practices. Acknowledging the negative effects of climate change on the western Joshua tree, we urge the Commission to formally designate this iconic species as "threatened" under CESA.

Four of the five peer reviews were conducted by reviewers who have studied *Yucca brevifolia*, and all four disagreed with either the Department of Fish and Wildlife's recommendation or the rationale behind the recommendation. The consensus of scientific evidence supports the listing of the western Joshua tree as "threatened." Importantly, this is an opportunity for our policy makers to respond to climate change proactively by applying the protections of CESA to species like the western Joshua tree that are facing decline in the face of the current climate crisis.

One peer reviewer noted, "I am not convinced that the available evidence supports a

recommendation against designating the species as ‘threatened.’ Current threats to western Joshua trees in California, considered in combination, mean that the species has very real potential to ‘become an endangered species in the foreseeable future’ (again, per Fish and Game Code, sect. 2067).”¹

Ample scientific evidence demonstrates that the species meets the criteria for listing as “threatened.” Though there is some uncertainty regarding how western Joshua trees will be affected in the future by increasing temperatures and changes in the timing and abundance of precipitation, it is well documented that climate change is already impacting this species. Range wide studies² and studies in Joshua Tree National Park³ predict near complete loss of suitable habitat for this species due to climate change. The status review document cites recruitment from the 1990s as evidence that populations are stable, but this evidence conflicts with field data in published studies showing that certain Joshua tree populations have reached a mature age and are no longer producing offspring to keep the populations stable. Without proper protection and management other populations will also become extirpated in the foreseeable future.

This led another peer reviewer to conclude that, “many are already ‘evolutionarily extinct’ populations of only mature adults, with no successful recruitment. Others will be unless we do something.”⁴

Another factor influencing recruitment is the western Joshua tree’s sole pollinator, *Tegeticula synthetica*, a species of yucca moth. Western Joshua trees at the edge of their range at higher elevations produce fewer viable seeds than populations at intermediate elevations, probably due to lower numbers of pollinators.⁵ This suggests that as populations are pushed beyond the northernmost and highest elevation portions of their range, pollinators may not follow. The loss of this pollinator or a decoupling of the timing of flowering and the activity of yucca moths due to climate change would effectively halt the reproduction of the western Joshua tree.

The effects of fire and herbivory both have been shown to have a significant impact on

¹ Peer review comments of Jeremy Yoder.

² Cole, K.L., Ironside, K., Eischeid, J., Garfin, G., Duffy, P.B. and Toney, C., 2011. Past and ongoing shifts in Joshua tree distribution support future modeled range contraction. *Ecological Applications*, vol. 21, no. 1, pp.137-149.

³ Barrows, C.W. and Murphy-Mariscal, M.L., 2012. Modeling impacts of climate change on Joshua trees at their southern boundary: how scale impacts predictions. *Biological Conservation*, vol. 152, pp.29-36; Sweet, L.C., Green, T., Heintz, J.G., Frakes, N., Graver, N., Rangitsch, J.S., Rodgers, J.E., Heacox, S. and Barrows, C.W., 2019. Congruence between future distribution models and empirical data for an iconic species at Joshua Tree National Park. *Ecosphere*, vol. 10, no. 6, p.e02763.

⁴ Peer review comments of Cameron Barrows.

⁵ Harrower, J. and G. S. Gilbert. 2018. Context-dependent mutualisms in the Joshua tree–yucca moth system shift along a climate gradient. *Ecosphere* vol. 9, no 9, pp. e02439. 10.1002/ecs2.2439

the recruitment and recovery of western Joshua tree populations. Blackbrush and Joshua tree communities typically convert to plant communities dominated by non-native annuals after fire and the recruitment of native shrubs and Joshua trees can take decades, if this happens at all.⁶ Additionally, annual dominated ecosystems are more likely to have more frequent fires,⁷ which can leave seed banks empty if subsequent fires occur before native shrubs and Joshua trees reach reproductive maturity. Joshua tree seedlings are typically found in the protected understory of shrubs,⁸ called nurse plants. Studies on saguaro cactus, a species that also requires nurse plants for recruitment, show that recovery of nurse plants to the point that they can support seedlings can take ten years after the cessation of grazing.⁹ This is likely true for areas recovering from wildfire as well.

Using fossil records as an indicator of the expected speed at which the range of Joshua trees can change is not reliable, as evidence suggests that current warming is happening approximately ten times faster than the rate of ice-age-recovery warming.¹⁰ As a result, we are concerned that western Joshua tree's ability to adapt/move will be far outpaced by the current rate of environmental change. This is likely to lead to extirpation of western Joshua trees in many locations.

Over 40% of the western Joshua tree's range in California is on private land, of which only a small percentage is protected from development and land management/use that is inconsistent with conservation. Federal lands that are home to populations of western Joshua tree are subject to a variety of threats including off-road vehicle use, cattle grazing, the spread of invasive annual plants, electrical transmission line and gas pipeline construction and maintenance, and large-scale energy projects. These threats are expected to become amplified in the coming decades. However, if the species and its habitats are protected and active management to enhance recruitment, survival, and distribution is employed, these threats are not insurmountable. Listing the western Joshua tree as "threatened" will help to alleviate some of these threats.

⁶ Hughes, L.E., 2002. Is there recovery after fire, drought, and overgrazing?. *Rangelands Archives*, vol. 24, no. 4, pp. 26-30.

⁷ Davies, K.W., Wollstein, K., Dragt, B. and O'Connor, C., 2022, Grazing management to reduce wildfire risk in invasive annual grass prone sagebrush communities, *Rangelands*, <https://doi.org/10.1016/j.rala.2022.02.001>

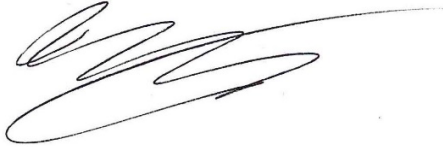
⁸ Brittingham, S.B., 1998. Facilitation of *Yucca brevifolia* recruitment by Mojave Desert shrubs (Doctoral dissertation, University of Nevada, Las Vegas).

⁹ Hall, J.A., S. Weinstein, and C.L. McIntyre. 2005. The Impacts of Livestock Grazing in the Sonoran Desert: A Literature Review and Synthesis. The Nature Conservancy in Arizona, Tucson

¹⁰ <https://climate.nasa.gov/evidence/>

Thank you for the opportunity to comment on this decision. Please don't hesitate to contact me if you have any questions.

Sincerely,

A handwritten signature in black ink, appearing to read 'Nick Jensen', with a long horizontal flourish extending to the right.

Nick Jensen, PhD
Conservation Program Director
California Native Plant Society
njensen@cnps.org



MOJAVE DESERT LAND TRUST

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Joshua Tree, California 92252

760.366.5440 • MDLT.org

California Fish and Game Commission

fgc@fgc.ca.gov

Subject: Western Joshua Tree Listing

June 2, 2022

Dear Members of the Fish and Game Commission,

We are writing to you to express our strong support for permanent listing of the western Joshua tree as a threatened species under CESA.

The Mojave Desert Land Trust is a 501c3 nonprofit organization dedicated to the conservation of the California desert ecosystem. Since our inception in 2006, we've protected more than 100,000 acres of desert habitat. We've conveyed a large percentage of those lands to national parks, wilderness areas and monuments, and currently hold more than 40,000 acres of desert with ecological significance, including riparian areas, springs, and wildlife corridors, much of which contain Joshua trees. As one of the largest land conservancies in the California desert, we have a significant stake in the fate of the western Joshua tree and its habitat.

In response to a large outpouring of support for greater protection for the western Joshua tree, we hosted a public rally in May that was enthusiastically attended by 110 concerned community members. Their messages of support and comments have been shared with the Commission.

As long-term landowners we clean up, restore, conduct research on and steward our lands, making us acutely tuned in to the threats the western Joshua tree faces. Climate change combined with drought, invasive species and human encroachment are taking a visible toll on the species.

In the last three years, we've had four major fires burn our lands, including one that ignited last week. This is a significant increase since our inception in 2006, and since fires have been reliably documented in the desert over the last 100 years. Proliferation of non-native plants has contributed to these increasingly frequent wildfires across desert lands, including devastating losses within western Joshua tree habitat. The convergence of increasing temperatures, invasive species, drought conditions, and a lengthier fire season are advancing both frequency and intensity of wildfires within Joshua tree habitat. This reality has forced us to take on a new role in fire restoration and management, and we are currently conducting research in conjunction with the BLM looking at Joshua tree recruitment post-fire in one of our parcels that recently burned near Joshua Tree National Park.

Climate change threats to the western Joshua tree are exacerbated by intense development pressure on western Joshua tree habitat that accounts for 40% of its range on private land. Residential, commercial, and more recently, poorly sited large-scale renewable energy projects have already impacted thousands of acres of western Joshua tree woodlands with no indications of decline. Indeed, as the popularity of the desert region grows in leaps and bounds, with the western Joshua tree as a signature species and tourist attractant, we can anticipate ever expanding development pressure. Development projections are particularly pronounced in the southern range

region. EPA's Integrated Climate and Land-Use Scenarios (ICLUS) modeling tool projected that 41.6% of suitable habitat in the southern habitat region would be lost to housing development by 2095.

MDLT's position in support of permanent protection for the western Joshua tree does not exist within a conservation-focused bubble. We recognize the economic context in which protections take place. The real estate market in the Joshua tree region has exploded. According to Redfin, home values in the Joshua Tree community skyrocketed by almost 69% between 2020 and 2022, in part because of its proximity to the National Park and the growing AirBnB market. The popularity of California's desert region has translated into clear beneficial economic metrics. Employment opportunities often cited by development projects tend to be short-term and limited, whereas tourism employment sparked by the public's interest in the western Joshua tree is expansive and long-term. In 2018, 2.9 million visitors to Joshua Tree National Park created a total economic benefit to the local region of \$196 million – that's \$68 million more than in 2015. For the local communities adjacent to the park, 1,823 jobs were related to Joshua Tree National Park visitation.

The western Joshua tree is much more than a spectacular desert plant. The identities of gateway communities to Joshua Tree National Park and much of the California desert region are intrinsically tied to the existence and perpetuity of this iconic plant. Visitors are attracted from around the world. In 2020, Joshua Tree National Park was the most visited park in the state of California, outperforming even Yosemite National Park – and the 10th most visited park in the entire United States.

The merging of conservation interests with development/economic considerations has the opportunity to be mutually beneficial. Proven success stories abound within Regional Habitat Conservation Plans and Natural Community Conservation Plans which allow for smart growth that streamlines development and permitting: this is the win-win situation we hope to see once permanent listing is secured.

The scientific community specializing in western Joshua tree research has left little doubt that the western Joshua tree is at tremendous risk, as noted in their peer reviews of the CDFW recommendation. We need our leaders to take bold action now if we are going to be able to protect the desert ecosystem and its ability to function for the biodiversity it contains, including us. As a keystone species, the western Joshua tree is the glue that holds the entire ecosystem together. Please consider the weight of this moment for the western Joshua tree, and for the ongoing conservation of the desert for generations to come.

We thank the Commission members for their integrity and leadership in voting for interim protections for the western Joshua tree during the 18-month review period. We hope that California can serve as a national example by voting for permanent listing of the western Joshua tree as a threatened species under CESA.

Sincerely,

A handwritten signature in black ink, appearing to read "K Herbinson". The signature is fluid and cursive, with a long horizontal stroke at the end.A handwritten signature in black ink, appearing to read "C Hanford". The signature is more compact and stylized than the one above, with a prominent loop at the end.

Kelly Herbinson
Cody Hanford
Joint Executive Directors
Mojave Desert Land Trust

RECEIVED
CALIFORNIA
FISH AND GAME
COMMISSION

Attention: Fish and Game Commission:

2022 MAY 11 AM 8:21

Hello. My name is Wendy Schiff, and I live in Palmdale, California, at the western end of the Mojave Desert. I've lived here most of my life and have witnessed the ongoing destruction of the Joshua tree and native desert habitat for over 30 years. Our desert land and Joshua trees are being torn down and clear-cut at record speed, all in the name of endless real estate development, industrial projects, strip malls, solar farms and wind farms. It has caused a massive scale die off in the desert, and it is absolutely heartbreaking to witness. In addition to large-scale destruction of the Joshua tree, other forms of desert plants and animals are disappearing from where I live as well; namely, quail, cottontail rabbits, jackrabbits, Fremont Cottonwood trees, Juniper trees and sagebrush.

Cities are finding loopholes in order to get around protective laws for Joshua trees by allowing companies to use "emergency take" permits to clear the land legally. These emergency take permits are being taken advantage of in the worst and most widespread way where I live and present an ongoing threat to the land. Cities claim to transplant the trees, but 60% die after transplantation.

In addition to killing the Joshua trees, animals will die off permanently in certain areas due to ongoing habitat destruction and land fragmentation. Ecosystems should be protected and not destroyed in the name of greed - in the ongoing quest to build tract homes, strip malls and solar farms.

Drought and fires are another major threat to the Joshua tree. I hike often, and constantly come across areas of burnt down Joshua trees. It is a sad sight. They need a chance to replenish their population. Many Joshua trees that have been burned and appear to be dead can possibly regenerate after 20 years or so. I've seen it with my own eyes.

If protections are lifted, Phase 1 of the massive, large-scale destruction of our deserts will begin immediately, mostly in the name of solar farms producing electricity for humans. This is incredibly self-serving and greedy on the part of human beings. From what I understand, the land traps carbon underground, but all of that carbon gets kicked back up into the atmosphere when land is developed. This makes pollution even worse and hurts the environment more. The government should require commercial buildings in the city to have solar panels on the roofs, and the government should also pay for it since they are pushing it so hard, and the number of solar farms allowed to be constructed should be limited by law.

Solar farms are a scam in the sense that they are not capable of providing electricity to the vast majority of people unless we clear upwards of 70% of natural desert land and replace it all with solar panels. What an awful idea for the future! What's more, a new form of technology will surely come along and replace solar panels in about five to ten years (think of a solar "prism"), but, by then, the damage will be done by the solar farm industry. The land will be dead and bare. I have serious questions as to why the California Department of Fish and Wildlife deemed that the Joshua tree should not be protected a week before the Commission is holding a meeting

regarding Joshua trees being potentially listed as an endangered species. It seems rather odd that an agency serving to protect nature is giving the green light to have it destroyed. I wonder who could have influenced their decision in that regard.

Americans have a long history of looking at land through the lens of how it can benefit them and what kind of resources they can derive from it. For example, how many crops someone can grow on several acres, how much money can be made from logging the trees and selling the wood, the sale of luxury fur coats, etc. But a new way of thinking is emerging in various countries in which people are beginning to believe that nature has its own rights and has a right to exist independent of what type of resources it can offer people. It is called the "Rights of Nature" movement, and some countries are beginning to grant certain natural areas the right to be left alone and untouched. However, when it comes to resources, perhaps the Joshua tree could provide us with a needed resource in the future, like medicine. There's a popular chemo drug called Taxol, which is derived from the bark of the Pacific yew tree.

In summary, I would urge you to think about everything I've stated in this letter. I am just an everyday layperson who cannot make any money from the development of open land. I want to preserve the Joshua tree and its habitat and the animals that utilize it from the goodness of my heart and for future generations to experience it. People love the desert and this iconic tree. Think of the popularity of Palm Springs and Joshua Tree State Park. Please put this ancient, mysterious and well-loved tree on the endangered species list permanently. It is the right thing to do.

Thank you very much for your time.

Sincerely,
Wendy Schiff



Fish and Game Commission
California Department of Transportation
Conference Room 1.040 (1st Floor)
100 S. Main Street
Los Angeles, CA 90012

RE: Agenda Item 5: Western Joshua tree
June 15, 2022

Dear Commissioners,

I am writing this statement as testament on the necessity for state protection of the Western Joshua tree (*Yucca brevifolia*). I have lived in San Bernardino County most of my adult life, including 17 years in the Mojave Desert communities of Victorville and Hesperia. As a registered voter and former resident, I have witnessed vandalism, negligence, and the destructive process of urban development in Hesperia for quite some time.

In 2016, I began to photographically document the mistreatment of Western Joshua trees that were under the protection of the county of San Bernardino and the city of Hesperia. My images provide evidence that the laws in place at the county and city level are not enough to protect the Western Joshua tree from systematic eradication due to urban development. The following pages contain images that provide support on why state protection is needed.

Image 1



Vacant lot, east of hwy. 395, north of Poplar St, Hesperia, CA. 2017. Vandalized tree cut down. No construction in the area at the time.

Image 2



South of 11510 Fashion Ct. Hesperia. 2018. Frozen Joshua tree from sprinklers. Water is often applied to help loosen the soil before the development of a construction site. Tree was not saved. Lot is still vacant as of May 27,2022.

Image 3



Adjacent to 8770 Caliente Rd, Oak Hills, CA. 2018. Pile of uprooted Joshua trees due to construction. No trees were saved.

Image 4



8770 Caliente Rd, Oak Hills, CA. 2018. Another pile of uprooted Joshua trees due to construction. No trees were saved.

Images 5 and 6



Joshua tree 164, South of 11412 Fashion Ct. Hesperia, CA. 2019. For monitoring and protection, local officials tag trees before the development of a specific site for construction.

Images 7, 8, 9



Even though it was tagged, Joshua tree 164 was broken at the trunk base and tossed into the surrounding juniper shrub during the development of vacant lot, south of 11412 Fashion Ct. Hesperia, CA. 2019. Tree was destroyed. Lot still vacant as of May 27, 2022.

With the presented photographic evidence, city and county officials are not capable of ensuring that Western Joshua trees will be protected under their supervision. Without state protection, more Joshua trees will be lost due to negligence and urban development. With the effects of climate change, increased fire behavior, and a hotter, drier climate already taking their toll; urban development could be the biggest contribution to the systematic eradication of an endemic species unique to the Mojave Desert. I implore the State of California Fish and Game Commission to act now and protect the Western Joshua tree species before it is too late.

Thank you.

Sincerely,

Frederick James Brashear Jr, M.F.A.
Artist & Educator

From: Mitch Miller <[REDACTED]>
Sent: Wednesday, May 25, 2022 11:54 AM
To: FGC
Subject: Protection for Western Joshua Trees

Categories: Exhibit

Ms. Samantha Murray, President California Fish and Game Commission

I am a Joshua Tree-based photographer. I've explored Joshua Tree National Monument/Park since Memorial Day 1980. My wife and I were married in the Park in 1999, moved here in 2017 and have seen changes. There are more drying, dying, withering, and generally sickly-looking Joshua trees as a result of climate change-induced drought and air pollution.

As an art show vendor, I've seen many travelers enjoy my photographs of pinyon pine, ocotillo, cholla, rocks, smoke trees and yucca. But it is the iconic Joshua tree in which they are most interested. Many, many people, from the maintenance personnel at the national park, to the barista at the local coffee shop, local bakers, convenience store cashiers, artists, restaurant servers and more rely on that very interest in this unique plant for their livelihood.

But despite the Joshua tree's amazing survivability it is under constant threat from climate change, drought, vandalism and development. If we can't protect this tourism magnet for its own sake (and we should), we must protect it for our own economic survival.

They are gravitating towards "refugios" on north-facing slopes at higher elevations, but that's a slow process and visitors won't see that. If we don't act now to protect Joshua trees we might as well change the name of the park to Creosote Bush National Park, and that doesn't have quite the same allure.

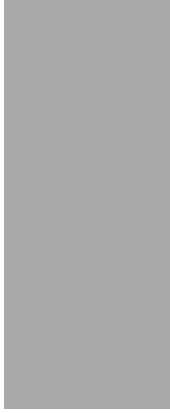
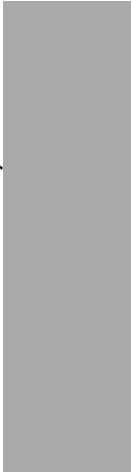
Please act now to give Joshua trees across their much-shrunk range the maximum protection possible under state law.

Thank You

Mitch Miller
Joshua Tree, CA

SAVE THE WESTERN JOSHUA TREE!

Write a message for the California Fish and Game Commission here. It will be sent to the Commission ahead of its June 15 and 16 meeting.

Name/Address	Personal Comment
JE. TUCKER 	These trees need our protection <u>NOW!</u>
JASON DUDLEY 	Proactively Save Our STREES Before it's too LATE!!!
MATTHEW CASE 	THE WESTERN JOSHUA TREE WILL NOT SURVIVE THE CENTURY WITHOUT OUR HELP. DO THE RIGHT THING FOR OUR AREA'S FUTURE AND LIST THE WJTT.
NEDA MCCULLOCH 	Please Save the Joshua Tree!

SAVE THE WESTERN JOSHUA TREE!

Write a message for the California Fish and Game Commission here. It will be sent to the Commission ahead of its June 15 and 16 meeting.

Brenda Estrella

As a Mojave Resident for over half my life, (since 2)
I have personally witnessed the horrible
decline of western Joshua Trees. It is very
horribly sad. The beauty of this Desert and the
wilderness of nature should be protected!!!

The world is at a Climate Crisis that
will take effect on the younger generations to come. We need
to think about all our actions and
the thing we can do to improve the
future. Please understand we the
people are noticing what is happening.
You can be the first state to
improve the wild habitats that
make California, CALIFORNIA.

Please be the First 1st
to make this

Thank you.
Cheryl Branda

SAVE THE WESTERN JOSHUA TREE!

Write a message for the California Fish and Game Commission here. It will be sent to the Commission ahead of its June 15 and 16 meeting.

Name/Address Joseph Bottari [Redacted Address]	Personal Comment Climate change is hurting these ancient trees please save them -
Jon Bottari [Redacted Address]	Please save our Joshua trees
Marjaret Strachan [Redacted Address]	Please protect the Joshua Trees. Allowing towns to rubber stamp take permits violates statute.
Noel Rhode [Redacted Address]	PLEASE PROTECT OUR PRECIOUS JOSHUA TREES

SAVE THE WESTERN JOSHUA TREE!

Write a message for the California Fish and Game Commission here. It will be sent to the Commission ahead of its June 15 and 16 meeting.

Name/Address	Personal Comment
ARCH McCULLOCH 	CLIMATE CHANGE IS IMPACTING THE ENTIRE DESERT. PROTECTING WESTERN JOSHUA TREES HABITAT WILL PROTECT MANY OTHER SPECIES OF PLANTS AND ANIMALS.
VERA TOPINKA 	Our climate change is affecting the entire desert. we <u>need</u> to help the joshua trees, the <u>western</u> joshua trees! ☺
Sarah Kennington 	
Noreen Lazor 	Please Save our Joshua trees

SAVE THE WESTERN JOSHUA TREE!

Write a message for the California Fish and Game Commission here. It will be sent to the Commission ahead of its June 15 and 16 meeting.

Debra Davis

I read the 158 page Joshua Tree report and it has only has so many mixed opinions, and different points of view. Reports about the survival of the Joshua trees. How could you possibly make

any decision based on conflicting scientific reports? I think it is obvious it is getting hotter and that the Joshua trees and all plant species are at greater risk of extinction. If anything the status should remain "threatened" to continue the study as there appears no conclusive results. Please save the Joshua trees.

SAVE THE WESTERN JOSHUA TREE!

Write a message for the California Fish and Game Commission here. It will be sent to the Commission ahead of its June 15 and 16 meeting.

Name/Address	Personal Comment
4/1/08 Gordon	Development is not halted to save the Joshua Tree. There is plenty of desert without Joshua Trees where development can happen without destroying the Joshua Trees. Please save the Joshua Trees.
Crystal Whelan	Please save our iconic prehistoric Joshua Tree. Be on the right side of history!
Doria Greenland	Change the distance that homes can be built to enable housing & Preserve the Joshua Trees. Protect our economy (Residents) need these trees.
DAMARIS ZATARAIN	THE WESTERN JOSHUA TREE IS IN DANGER OF EXTINCTION DUE TO DEVELOPMENTS THAT ARE BEING BUILT AROUND AN IDEA THAT THE TREES WILL STICK AROUND FOR PEOPLE + THE ENVIRONMENT TO ENJOY. PLEASE HELP US.

SAVE THE WESTERN JOSHUA TREE

Write a message for the California Fish and Game Commission here. It will be sent to the Commission ahead of its June 15 and 16 meeting.

Name/Address	Personal Comment
CONRAD LANGER 	PLEASE END OF THE SIDE OF CALIFORNIA. THE WORLD LOVES JOSHUA TREE + ITS GREAT SCENICITY + JAVELIN.

June 2, 2022

Samantha Murray, President
California Fish and Game Commission
1416 Ninth Street, Suite 1320
Sacramento, CA 95814

Submitted via email to fgc@fgc.ca.gov

RE: Petition to list western Joshua Tree as threatened under the California Endangered Species Act

Dear President Murray and Commissioners:

Thank you for the opportunity to submit comments regarding your upcoming decision to list the western Joshua tree (*Yucca brevifolia*) as “threatened” under the California Endangered Species Act.

Over the last several years, my botanical research and work required frequent visits to the Inyo Mountains, a site of potential climate refugia for *Yucca brevifolia* identified by Cole et al. 2011. As far as I am aware, there relatively little demography data on the northern populations of *Y. brevifolia* which mostly occur on federal lands. While my hope is that quantitative demography data will eventually be available for these northern populations, observational data may be of use in the Commission’s upcoming vote.

Following the onset of exceptional drought in Inyo County (2021 to present), I observed a rapid increase in mortality of many large Joshua trees due to small mammal herbivory (see photos below). The mortality appears to be localized in the southern portion of Lee Flat (Death Valley National Park) and has impacted several hundred mature trees in the area I observed. While the species is widespread and abundant in the northern part of its range, I find the rapid demise of mature trees at relatively high elevations (ca. 5,500’) cause for concern. These biotic interactions are not accounted for in the models predicting the future range of *Y. brevifolia* cited in the Center for Biological Diversity’s listing petition.

Given the likelihood that very few climate refugia exist in California (Cole et al. 2011; Sweet et al. 2019), it follows that protecting habitat in these areas will be necessary for the continuation of the species. However, if *Y. brevifolia* loses protective status, it is possible that some of these places will be lost to development, leaving the refugia that remain more vulnerable to stochastic loss. For instance, candidacy status was cited as one of the reasons the Bureau of Land Management required an Environmental Impact Statement (see attached letter from BLM) in response to a mining company’s request to develop an exploratory drilling project in the southern Inyo Mountains – a high-elevation climate refugia that could be lost forever if the mining project continues on its trajectory leading to an open-pit gold mine.

I urge you to consider the long-standing impacts the listing decision will have on the future viability of *Yucca brevifolia* and the integrity of its habitat in the Mojave Desert.

Respectfully,

A stylized, handwritten signature in black ink, appearing to be 'MJ' with a long, sweeping horizontal stroke extending to the right.

Maria Jesus
Bishop, CA

Attachment 1: Photos of Joshua tree herbivory and mortality at Lee Flat, Death Valley National Park,





United States Department of the Interior

BUREAU OF LAND MANAGEMENT

Ridgecrest Field Office
300 S. Richmond Road
Ridgecrest, CA 93555
www.ca.blm.gov/ridgecrest



In Reply Refer To:
CACA-56495
3809(P)
LLCAD05000.51

March 09, 2022

Mojave Precious Metals, Inc.
Attn: Steven Swatton
1020-800 Pender Street W
Vancouver, BC V6C 2VC

Dear Mr. Swatton,

After a thorough review of the Plan of Operations modification that you submitted on behalf of Mojave Precious Metals, Inc. (MPM), the BLM has determined that it will analyze the potential impacts in accordance with the National Environmental Policy Act by preparing an Environmental Impact Statement (EIS). The EIS will analyze the following resources in detail: wildlife, vegetation, special status species, noxious weeds, cultural and Native American resources, special land use designations, recreation, water, soil, and visual resources (and potentially transportation and access). MPM's original Plan of Operations was analyzed in an Environmental Assessment (EA) and the BLM issued its Finding of No Significant Impact (FONSI) and Decision Record in May 2018. In that decision, the BLM approved the alternative that authorized access via helicopter only and created a surface disturbance of 0.2 acres. The current proposed plan modification calls for access via overland travel and road construction and could disturb over 12 acres. Based on the resource concerns expressed by the public, Tribes, and agencies during scoping, the increased surface disturbance of the proposed plan modification, and the increase to 30 drill locations with a total of 120 hole, the BLM has determined that an EIS is warranted.

The BLM also considered the following factors when determining that it will prepare an EIS for the proposed plan modification:

1. The Joshua Tree is now a candidate species under the California Endangered Species Act.
2. The area is considered to be Lands with Wilderness Characteristics (WIU #124-1).
3. Red Ochre clay for which the Timbisha Tribe is named was unearthed during the first drilling project.
4. The largest known population of Inyo thread plant (*Nemacladus inyoensis*) in the southern Inyo Mountains was observed growing in and adjacent to the reclaimed access road. This plant species is new to science as of 2020 and is currently undergoing a California Rare Plant Rank status review (proposed status 1B.2).

5. On a field visit to engage in government-to-government consultation, Tribes told the BLM that the proposed plan modification could impact the holistic value of the area. Tribes explained that the area contains historic trails and sites where men of the tribe would gather for various reasons.
6. Under the Desert Renewable Energy Conservation Plan, portions of the area are designated as Areas of Critical Environmental Concern and California Desert National Conservation Lands.
7. The proposed plan modification would cause an increase in water use, which may result in potential impacts on quantity and quality to water supplies, seeps, springs, streams or water reservoirs. There could also be indirect impacts from water withdrawal, transport, and storage on air quality, dust generation, and traffic.

If you have any questions, please contact me at (760)384-5400 or Randy Porter at (760)384-5452 and rporter@blm.gov.

Respectfully,

CARL
SYMONS

Digitally signed by CARL
SYMONS
Date: 2022.03.09 10:16:34
-08'00'

Carl Symons
Ridgecrest Field Manager